

COMP0008: Computer Architecture & Concurrency – CW1

Dear students,

These are the solutions for CW1. We strongly recommend that you try solving the questions yourself before looking at the answers.

COMP0008 staff.

Solution:

1. Let $x = 13_4$ and $y = 11_{16}$. How many digits $(x \times y)$ has in base 64?

We have that $y = 11_{16} = 101_4$. Thus $x \times y = 303_4 + 1010_4 = 1313_4$. We have that $(10^3)_4 < 1313_4 < (10^6)_4 = 10_{64}$, and thus $x \times y$ has a two base-64 digits. This uses the fact that $4^3 = 64$ and thus every base-64 digit is equivalent to three base-four digits.

An alternative solution, which many of you may have used is to convert to decimal: $x = 1 \cdot 4^1 + 3 \cdot 4^0 = 7_{10}$ and $y = 10 \cdot 16^1 + 1 \cdot 16^0 = 17_{10}$ which gives $(x \times y) = 119_{10}$. Since $64 < 119 < 64^2$, the product requires two digits.

The takeaway: *it is sometimes easier to do computation in bases other than 10.*

2. Let x, y be 32-bit integers. How many bits are guaranteed to be enough for representing $(x \times y)$ for any values of x, y ?

64-bits. This is because x, y take values that can be at most $2^{32} - 1$ and thus $2^{63} < (2^{31} - 1)^2 < 2^{64}$, which means that 64 bits are enough but 63 are not.

The takeaway: *the product of numbers may require more bits. Trivia: What does it mean for multiplication in MIPS32 (where all registers are 32-bit long)?*

3. Suppose that x and y both have d digits (without leading zeros) in base b . How many digits will $(x \times y)$ have in base b^2 ? Give the smallest feasible interval $[l, h]$.

We have that $x < b^d, y < b^d$ have thus $(x \times y) < b^{2d}$ and hence needs no more than $2d$ digits in base b and thus d digits in base b^2 .

Similarly, if both are non-zeros, $x \geq b^{d-1}$ and $y \geq b^{d-1}$ which means that $(x \times y) \geq b^{2d-2}$ and thus it requires at least $2d - 1$ in base b and then at least $\left\lceil \frac{2d-1}{2} \right\rceil = d$ digits will be needed in base b^2 . Finally, if $(x \times y) = 0$ then a single digit is needed.

To conclude, **exactly** d digits will be required to represent the product in base b^2 .

The takeaway: *the product of numbers may require more bits. Trivia: What does it mean for multiplication in MIPS32 (where all registers are 32-bit long)?*

4. What is the 2's complement of 0xBADC0C0A? Calculate it without going through binary. We complement each digit: 0x4523F3F5 and then add 1 to get 0x4523F3F6.

This follows the next cool trick:

If we take the *bitwise-not* of an N -bit (in this example, $N = 32$) binary number x and denote it by x' , then $(x + x' + 1) = 2^N$ (i.e., $x + x'$ is the largest number representable using N bits). This means that the 2's complement of x is $(x' + 1)$, i.e., we can simply "flip" each bit in x and then add 1. Moving to hexadecimal numbers, we can see that by replacing each digit z by $(15 - z)$ and adding 1 at the end, we can derive the complement.

The takeaway: *find your preferred way of calculating 2's complement of numbers quickly, it will appear in more modules (and depending on what you do, in your future work) than you currently imagine!*